**Annual Report - General Order No. R5-2007-0035**
*Reporting period 01/01/2023 to 12/31/2023.*

DAIRY FACILITY INFORMATION

**A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY:** Fire-Lake Jerseys

Physical address of dairy:

| Number and Street | City | County | Zip Code |
|---|---|---|---|
| 11301 W Bradbury RD | Turlock | Stanislaus | 95380 |

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1950

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X058-X025-X004-XXXX X058-X025-X005-XXXX

**B. OPERATORS**

Sanders, Cathy

Operator name: Sanders, Cathy

Telephone no.: (209) 632-2924 Landline (209) 678-2670 Cellular

| Mailing Address Number and Street | City | State | Zip Code |
|---|---|---|---|
| P.O. Box 1024 | Hilmar | CA | 95324 |

**This operator is responsible for paying permit fees.**

**C. OWNERS**

Sanders, Scott & Sharlee

Legal owner name: Sanders, Scott & Sharlee

Telephone no.: (209) 632-2924 Landline (209) 541-9097 Cellular

| Mailing Address Number and Street | City | State | Zip Code |
|---|---|---|---|
| P.O. Box 459 | Hilmar | CA | 95324 |

**Annual Report - General Order No. R5-2007-0035**
*Reporting period 01/01/2023 to 12/31/2023.*

AVAILABLE NUTRIENTS

## A. HERD INFORMATION

| | Milk Cows | Dry Cows | Bred Heifers (15-24 mo.) | Heifers (7-14 mo. to breeding) | Calves (4-6 mo.) | Calves (0-3 mo.) |
|---|---|---|---|---|---|---|
| Number open confinement | 22 | 24 | 112 | 40 | 22 | 55 |
| Number under roof | 216 | 0 | 0 | 22 | 0 | 0 |
| Maximum number | 240 | 24 | 112 | 62 | 22 | 55 |
| Average number | 236 | 24 | 112 | 62 | 22 | 55 |
| Avg live weight (lbs) | 1,100 | 1,250 | 750 | 525 | | |

Predominant milk cow breed: Jersey

Average milk production: 54 *pounds per cow per day*

## B. MANURE GENERATED

Total manure excreted by the herd: 7,836.47 *tons per reporting period*

Total nitrogen from manure: 98,182.88 *lbs per reporting period*

After ammonia losses (30% loss applied): 68,728.02 *lbs per reporting period*

Total phosphorus from manure: 15,985.03 *lbs per reporting period*

Total potassium from manure: 42,715.35 *lbs per reporting period*

Total salt from manure: 116,639.40 *lbs per reporting period*

## C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 6,687,210 *gallons*

Total nitrogen generated: 17,175.00 *lbs*

Total phosphorus generated: 4,753.90 *lbs*

Total potassium generated: 30,620.55 *lbs*

Total salt generated: 232,949.94 *lbs*

| | | |
|---|---|---|
| | 6,687,210 | *gallons applied* |
| + | 0 | *gallons exported* |
| - | 0 | *gallons imported* |
| = | 6,687,210 | *gallons generated* |

## D. FRESH WATER SOURCES

| Source Description | Type |
|---|---|
| Domestic Well - 4-1 Facility | Ground water |
| Domestic Well - 4-2 House | Ground water |
| Irrigation Water - Canal | Surface water |

## E. SUBSURFACE (TILE) DRAINAGE SOURCES

*No subsurface (tile) drainage sources entered.*

## F. NUTRIENT IMPORTS

*No dry manure nutrient imports entered.*

*No process wastewater nutrient imports entered.*

*No commercial or other nutrient imports entered.*

## G. NUTRIENT EXPORTS

| Date | Material type | Quantity | Reporting basis | Moisture (%) | Density (lbs/cu ft) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|---|
| 05/24/2023 | Corral solids | 900.00 *ton* | Dry-weight | 8.7 | | 23,200.00 | 8,300.00 | 22,900.00 | | 0.00 |

*No liquid nutrient exports entered.*

| Material type | Total N (lbs) | Total P (lbs) | Total K (lbs) | Total salt (lbs) |
|---|---|---|---|---|
| Dry manure | 38,126.88 | 13,640.22 | 37,633.86 | 0.00 |
| Process wastewater | 0.00 | 0.00 | 0.00 | 0.00 |
| Total exports for all materials | 38,126.88 | 13,640.22 | 37,633.86 | 0.00 |

APPLICATION AREA

## A. LIST OF LAND APPLICATION AREAS

| Field name | Controlled acres | Cropable acres | Total harvests | Type of waste applied | Parcel number |
|---|---|---|---|---|---|
| Field 2 | 10 | 10 | 2 | process wastewater | X058-X025-X004-XXXX |
| Field 3 | 5 | 5 | 2 | process wastewater | X058-X025-X005-XXXX |
| West 20 | 20 | 20 | 2 | process wastewater | X058-X025-X005-XXXX |
| Totals for areas that were used for application | 35 | 35 | 6 | | |
| Totals for areas that were not used for application | | | | | |
| Land application area totals | 35 | 35 | 6 | | |

## B. CROPS AND HARVESTS

### Field 2

Field name: Field 2

#### 11/15/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 10 Plant date: 11/15/2022

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 05/05/2023 | 159.30 *ton* | Dry-weight | | 80.3 | 17,400.00 | 3,100.00 | 24,800.00 | | 11.40 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 16.00 | 160.00 | 25.60 | 132.80 | 0.00 |
| Total actual harvest content | 15.93 | 109.21 | 19.46 | 155.66 | 715.51 |

#### 05/28/2023: Corn, silage

Crop: Corn, silage Acres planted: 10 Plant date: 05/28/2023

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 09/18/2023 | 251.40 *ton* | Dry-weight | | 66.7 | 16,800.00 | 3,500.00 | 16,200.00 | | 6.70 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 30.00 | 240.00 | 45.00 | 198.00 | 0.00 |
| Total actual harvest content | 25.14 | 281.29 | 58.60 | 271.24 | 1,121.80 |

## Field 3

Field name: Field 3

### 11/15/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 5 Plant date: 11/15/2022

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 05/05/2023 | 82.30 *ton* | Dry-weight | | 80.3 | 17,400.00 | 3,100.00 | 24,800.00 | | 11.40 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 16.00 | 160.00 | 25.60 | 132.80 | 0.00 |
| Total actual harvest content | 16.46 | 112.84 | 20.10 | 160.83 | 739.32 |

### 05/28/2023: Corn, silage

Crop: Corn, silage Acres planted: 5 Plant date: 05/28/2023

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 09/18/2023 | 125.60 *ton* | Dry-weight | | 66.7 | 16,800.00 | 3,500.00 | 16,200.00 | | 6.70 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 30.00 | 240.00 | 45.00 | 198.00 | 0.00 |
| Total actual harvest content | 25.12 | 281.06 | 58.55 | 271.02 | 1,120.90 |

## West 20

Field name: West 20

### 11/15/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 20 Plant date: 11/15/2022

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 05/05/2023 | 319.30 *ton* | Dry-weight | | 79.7 | 17,100.00 | 2,900.00 | 25,500.00 | | 2.30 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 16.00 | 160.00 | 25.60 | 132.80 | 0.00 |
| Total actual harvest content | 15.97 | 110.84 | 18.80 | 165.29 | 149.08 |

West 20

05/28/2023: Corn, silage

Crop: Corn, silage Acres planted: 20 Plant date: 05/28/2023

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 09/18/2023 | 462.30 *ton* | Dry-weight | | 64.3 | 17,700.00 | 3,500.00 | 16,300.00 | | 6.70 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 30.00 | 240.00 | 45.00 | 198.00 | 0.00 |
| Total actual harvest content | 23.12 | 292.12 | 57.76 | 269.02 | 1,105.78 |

NUTRIENT BUDGET

## A. LAND APPLICATIONS

### Field 2 - 11/15/2022: Oats, silage-soft dough

Field name: Field 2

Crop: Oats, silage-soft dough — Plant date: 11/15/2022

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 11/04/2022 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 74.42 | 17.60 | 111.84 | 840.30 | 245,000.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 29.68 | 1,077,600.00 *gal* |
| Application event totals | | 74.42 | 17.60 | 111.84 | 869.98 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 03/10/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 71.38 | 16.88 | 107.27 | 806.00 | 235,000.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 27.82 | 1,010,250.00 *gal* |
| Application event totals | | 71.38 | 16.88 | 107.27 | 833.82 | |

### Field 2 - 05/28/2023: Corn, silage

Field name: Field 2

Crop: Corn, silage — Plant date: 05/28/2023

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 05/21/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 65.31 | 15.45 | 98.14 | 737.41 | 215,000.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 48.22 | 1,751,100.00 *gal* |
| Application event totals | | 65.31 | 15.45 | 98.14 | 785.63 | |

## Field 2 - 05/28/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/30/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 41.50 | 14.53 | 87.17 | 664.83 | 221,300.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 46.37 | 1,683,750.00 *gal* |
| Application event totals | | 41.50 | 14.53 | 87.17 | 711.20 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/11/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 46.37 | 1,683,750.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 46.37 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/23/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 59.09 | 16.79 | 117.81 | 908.36 | 219,900.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 49.15 | 1,784,775.00 *gal* |
| Application event totals | | 59.09 | 16.79 | 117.81 | 957.51 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/05/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 60.16 | 17.10 | 119.95 | 924.88 | 223,900.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 50.08 | 1,818,450.00 *gal* |
| Application event totals | | 60.16 | 17.10 | 119.95 | 974.96 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/22/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 60.30 | 17.13 | 120.22 | 926.95 | 224,400.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 40.80 | 1,481,700.00 *gal* |
| Application event totals | | 60.30 | 17.13 | 120.22 | 967.75 | |

## Field 2 - 05/28/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 09/02/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 60.08 | 17.07 | 119.79 | 923.64 | 223,600.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 43.59 | 1,582,725.00 *gal* |
| Application event totals | | 60.08 | 17.07 | 119.79 | 967.23 | |

| | | | | |
|---|---|---|---|---|
| 09/09/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 42.66 | 1,549,050.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 42.66 | |

## Field 3 - 11/15/2022: Oats, silage-soft dough

Field name: Field 3

Crop: Oats, silage-soft dough Plant date: 11/15/2022

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 11/04/2022 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 82.01 | 19.40 | 123.25 | 926.04 | 135,000.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 33.38 | 606,150.00 *gal* |
| Application event totals | | 82.01 | 19.40 | 123.25 | 959.43 | |

| | | | | |
|---|---|---|---|---|
| 03/09/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 63.79 | 15.09 | 95.86 | 720.26 | 105,000.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 25.97 | 471,450.00 *gal* |
| Application event totals | | 63.79 | 15.09 | 95.86 | 746.22 | |

## Field 3 - 05/28/2023: Corn, silage

## Field 3 - 05/28/2023: Corn, silage

Field name: Field 3

Crop: Corn, silage

Plant date: 05/28/2023

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 05/21/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 59.12 | 16.80 | 117.86 | 908.77 | 110,000.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 33.38 | 606,150.00 *gal* |
| Application event totals | | 59.12 | 16.80 | 117.86 | 942.16 | |

| | | | | |
|---|---|---|---|---|
| 06/30/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 58.79 | 16.71 | 117.22 | 903.81 | 109,400.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 29.68 | 538,800.00 *gal* |
| Application event totals | | 58.79 | 16.71 | 117.22 | 933.49 | |

| | | | | |
|---|---|---|---|---|
| 07/11/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 35.24 | 639,825.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 35.24 | |

| | | | | |
|---|---|---|---|---|
| 07/23/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 68.35 | 16.17 | 102.71 | 771.70 | 112,500.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 31.53 | 572,475.00 *gal* |
| Application event totals | | 68.35 | 16.17 | 102.71 | 803.23 | |

| | | | | |
|---|---|---|---|---|
| 08/05/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 51.86 | 14.74 | 103.40 | 797.24 | 96,500.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 48.22 | 875,550.00 *gal* |
| Application event totals | | 51.86 | 14.74 | 103.40 | 845.46 | |

## Field 3 - 05/28/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/22/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 53.53 | 15.21 | 106.72 | 822.85 | 99,600.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 35.24 | 639,825.00 *gal* |
| Application event totals | | 53.53 | 15.21 | 106.72 | 858.09 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 09/02/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 53.15 | 15.10 | 105.97 | 817.07 | 98,900.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 33.38 | 606,150.00 *gal* |
| Application event totals | | 53.15 | 15.10 | 105.97 | 850.45 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 09/09/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 46.37 | 841,875.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 46.37 | |

## West 20 - 11/15/2022: Oats, silage-soft dough

Field name: West 20

Crop: Oats, silage-soft dough

Plant date: 11/15/2022

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 11/04/2022 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 70.32 | 24.63 | 147.71 | 1,126.58 | 750,000.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 33.85 | 2,458,275.00 *gal* |
| Application event totals | | 70.32 | 24.63 | 147.71 | 1,160.42 | |

## West 20 - 11/15/2022: Oats, silage-soft dough

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 03/10/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 71.25 | 24.96 | 149.68 | 1,141.60 | 760,000.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 31.07 | 2,256,225.00 *gal* |
| Application event totals | | 71.25 | 24.96 | 149.68 | 1,172.66 | |

## West 20 - 05/28/2023: Corn, silage

Field name: West 20

Crop: Corn, silage

Plant date: 05/28/2023

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 05/21/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 63.11 | 14.93 | 94.83 | 712.54 | 415,500.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 37.09 | 2,694,000.00 *gal* |
| Application event totals | | 63.11 | 14.93 | 94.83 | 749.63 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/30/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 62.51 | 14.79 | 93.94 | 705.85 | 411,600.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 36.63 | 2,660,325.00 *gal* |
| Application event totals | | 62.51 | 14.79 | 93.94 | 742.48 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/11/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 36.17 | 2,626,650.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 36.17 | |

West 20 - 05/28/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/23/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 64.17 | 15.18 | 96.43 | 724.54 | 422,500.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 35.70 | 2,592,975.00 *gal* |
| Application event totals | | 64.17 | 15.18 | 96.43 | 760.25 | |

| | | | | |
|---|---|---|---|---|
| 08/05/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 63.87 | 15.11 | 95.98 | 721.13 | 420,510.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 36.63 | 2,660,325.00 *gal* |
| Application event totals | | 63.87 | 15.11 | 95.98 | 757.76 | |

| | | | | |
|---|---|---|---|---|
| 08/22/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 39.36 | 13.79 | 82.68 | 630.58 | 419,800.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 35.24 | 2,559,300.00 *gal* |
| Application event totals | | 39.36 | 13.79 | 82.68 | 665.82 | |

| | | | | |
|---|---|---|---|---|
| 09/02/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon Water | Process wastewater | 55.39 | 15.74 | 110.44 | 851.56 | 412,300.00 *gal* |
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 39.88 | 2,896,050.00 *gal* |
| Application event totals | | 55.39 | 15.74 | 110.44 | 891.44 | |

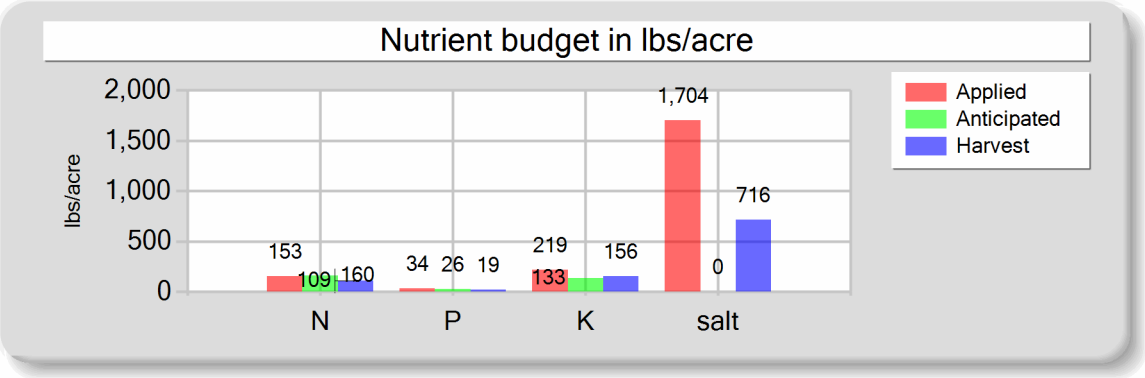
| | | | | |
|---|---|---|---|---|
| 09/09/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Irrigation Water - Canal | Surface water | 0.00 | 0.00 | 0.00 | 39.41 | 2,862,375.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 39.41 | |

## B. NUTRIENT BUDGET

Field 2 - 11/15/2022: Oats, silage-soft dough

Field name: Field 2 Crop: Oats, silage-soft dough Plant date: 11/15/2022




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 145.80 | 34.49 | 219.11 | 1,646.30 |
| Fresh water | 0.00 | 0.00 | 0.00 | 57.50 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 152.80 | 34.49 | 219.11 | 1,703.80 |
| Anticipated crop nutrient removal | 160.00 | 25.60 | 132.80 | 0.00 |
| Actual crop nutrient removal | 109.21 | 19.46 | 155.66 | 715.51 |
| Nutrient balance | 43.59 | 15.03 | 63.45 | 988.29 |
| Applied to removed ratio | 1.40 | 1.77 | 1.41 | 2.38 |

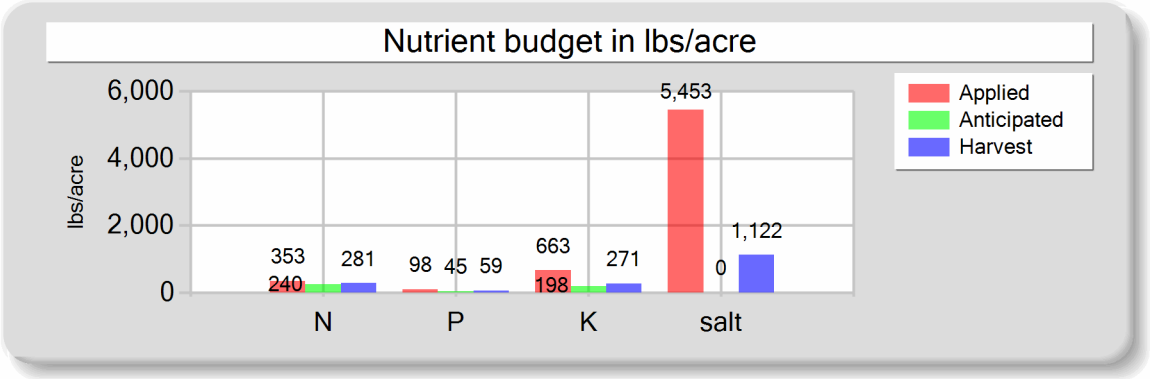
| Fresh water applied | |
|---|---|
| 2,087,850.00 | *gallons* |
| 76.89 | *acre-inches* |
| 7.69 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 480,000.00 | *gallons* |
| 17.68 | *acre-inches* |
| 1.77 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## Field 2 - 05/28/2023: Corn, silage

Field name: Field 2 Crop: Corn, silage Plant date: 05/28/2023




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 346.44 | 98.08 | 663.09 | 5,086.06 |
| Fresh water | 0.00 | 0.00 | 0.00 | 367.23 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 353.44 | 98.08 | 663.09 | 5,453.29 |
| Anticipated crop nutrient removal | 240.00 | 45.00 | 198.00 | 0.00 |
| Actual crop nutrient removal | 281.29 | 58.60 | 271.24 | 1,121.80 |
| Nutrient balance | 72.15 | 39.48 | 391.85 | 4,331.50 |
| Applied to removed ratio | 1.26 | 1.67 | 2.44 | 4.86 |

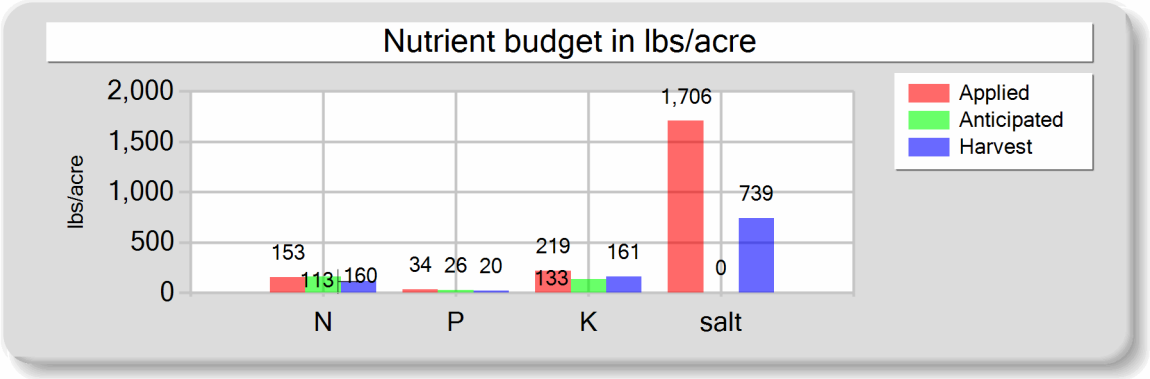
| Fresh water applied | |
|---|---|
| 13,335,300.00 | *gallons* |
| 491.09 | *acre-inches* |
| 49.11 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 1,328,100.00 | *gallons* |
| 48.91 | *acre-inches* |
| 4.89 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## Field 3 - 11/15/2022: Oats, silage-soft dough

Field name: Field 3 Crop: Oats, silage-soft dough Plant date: 11/15/2022




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 145.80 | 34.49 | 219.11 | 1,646.30 |
| Fresh water | 0.00 | 0.00 | 0.00 | 59.35 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 152.80 | 34.49 | 219.11 | 1,705.65 |
| Anticipated crop nutrient removal | 160.00 | 25.60 | 132.80 | 0.00 |
| Actual crop nutrient removal | 112.84 | 20.10 | 160.83 | 739.32 |
| Nutrient balance | 39.96 | 14.38 | 58.27 | 966.34 |
| Applied to removed ratio | 1.35 | 1.72 | 1.36 | 2.31 |

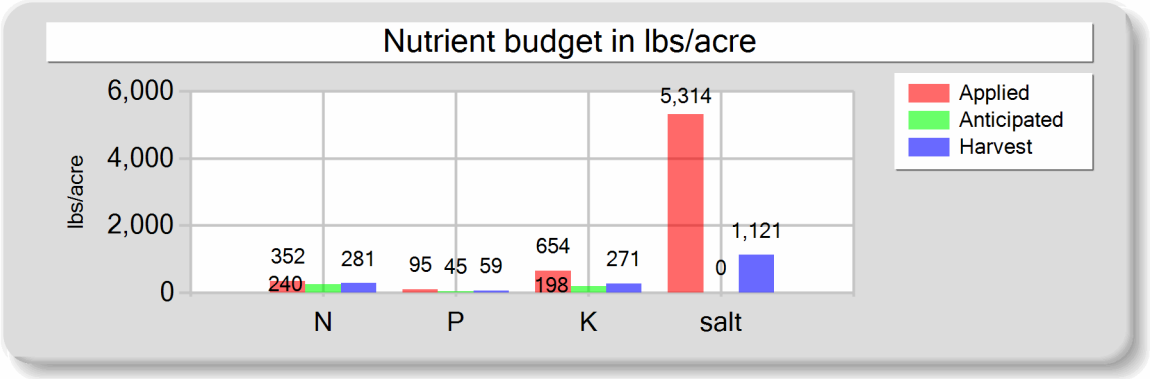
| Fresh water applied | |
|---|---|
| 1,077,600.00 | *gallons* |
| 39.68 | *acre-inches* |
| 7.94 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 240,000.00 | *gallons* |
| 8.84 | *acre-inches* |
| 1.77 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## Field 3 - 05/28/2023: Corn, silage

Field name: Field 3 Crop: Corn, silage Plant date: 05/28/2023




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 344.79 | 94.72 | 653.88 | 5,021.45 |
| Fresh water | 0.00 | 0.00 | 0.00 | 293.05 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 351.79 | 94.72 | 653.88 | 5,314.49 |
| Anticipated crop nutrient removal | 240.00 | 45.00 | 198.00 | 0.00 |
| Actual crop nutrient removal | 281.06 | 58.55 | 271.02 | 1,120.90 |
| Nutrient balance | 70.73 | 36.17 | 382.86 | 4,193.59 |
| Applied to removed ratio | 1.25 | 1.62 | 2.41 | 4.74 |

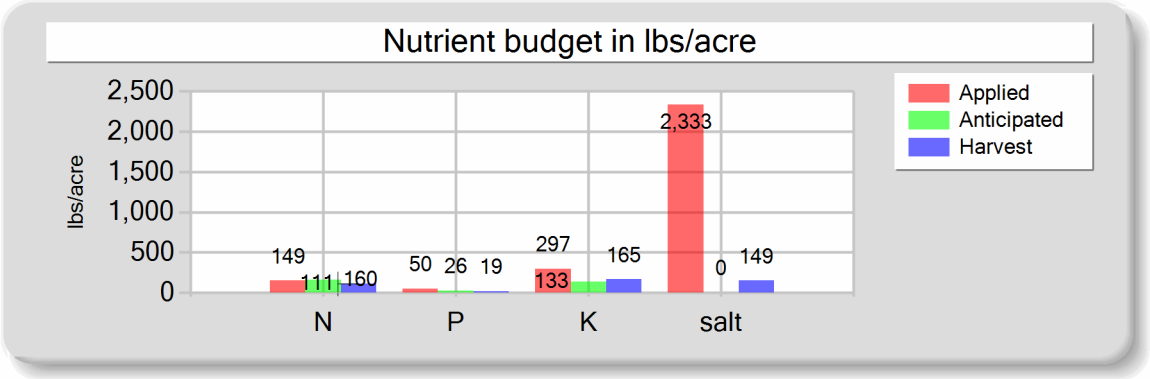
| Fresh water applied | |
|---|---|
| 5,320,650.00 | *gallons* |
| 195.94 | *acre-inches* |
| 39.19 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 626,900.00 | *gallons* |
| 23.09 | *acre-inches* |
| 4.62 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## West 20 - 11/15/2022: Oats, silage-soft dough

Field name: West 20 Crop: Oats, silage-soft dough Plant date: 11/15/2022




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 141.57 | 49.58 | 297.38 | 2,268.17 |
| Fresh water | 0.00 | 0.00 | 0.00 | 64.92 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 148.57 | 49.58 | 297.38 | 2,333.09 |
| Anticipated crop nutrient removal | 160.00 | 25.60 | 132.80 | 0.00 |
| Actual crop nutrient removal | 110.84 | 18.80 | 165.29 | 149.08 |
| Nutrient balance | 37.73 | 30.79 | 132.10 | 2,184.00 |
| Applied to removed ratio | 1.34 | 2.64 | 1.80 | 15.65 |

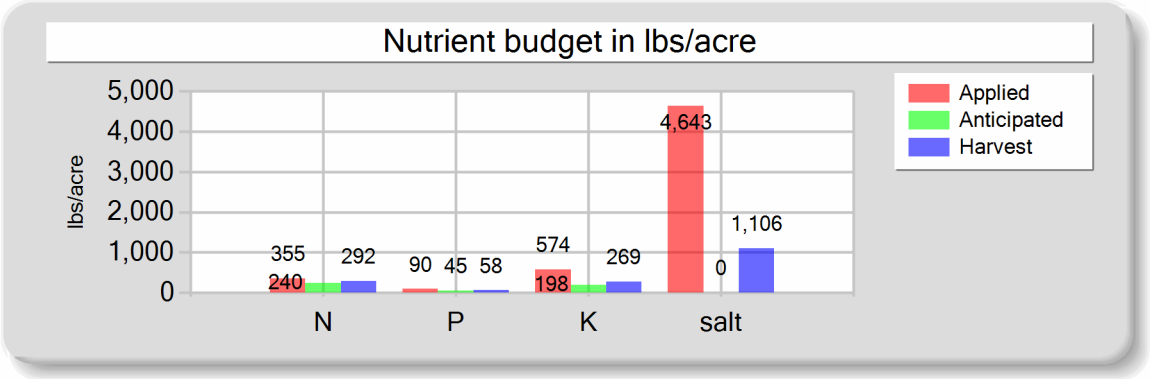
| Fresh water applied | |
|---|---|
| 4,714,500.00 | *gallons* |
| 173.62 | *acre-inches* |
| 8.68 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 1,510,000.00 | *gallons* |
| 55.61 | *acre-inches* |
| 2.78 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## West 20 - 05/28/2023: Corn, silage

Field name: West 20 Crop: Corn, silage Plant date: 05/28/2023




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 348.41 | 89.53 | 574.30 | 4,346.21 |
| Fresh water | 0.00 | 0.00 | 0.00 | 296.75 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 355.41 | 89.53 | 574.30 | 4,642.96 |
| Anticipated crop nutrient removal | 240.00 | 45.00 | 198.00 | 0.00 |
| Actual crop nutrient removal | 292.12 | 57.76 | 269.02 | 1,105.78 |
| Nutrient balance | 63.29 | 31.76 | 305.28 | 3,537.19 |
| Applied to removed ratio | 1.22 | 1.55 | 2.13 | 4.20 |

| Fresh water applied | |
|---|---|
| 21,552,000.00 | *gallons* |
| 793.69 | *acre-inches* |
| 39.68 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 2,502,210.00 | *gallons* |
| 92.15 | *acre-inches* |
| 4.61 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

NUTRIENT ANALYSES

## A. MANURE ANALYSES

### Solid Manure

Sample and source description: Solid Manure

Sample date: 05/24/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 4.2 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Calcium (mg/kg) | Magnesium (mg/kg) | Sodium (mg/kg) | Sulfur (mg/kg) | Chloride (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 9,000.00 | 3,900.00 | 10,600.00 | | | | | | | |
| **DL** | 100.00 | 100.00 | 100.00 | | | | | | | |

### Solid Manure

Sample and source description: Solid Manure

Sample date: 10/16/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 8.7 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Calcium (mg/kg) | Magnesium (mg/kg) | Sodium (mg/kg) | Sulfur (mg/kg) | Chloride (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 23,200.00 | 8,300.00 | 22,900.00 | | | | | | | |
| **DL** | 100.00 | 100.00 | 100.00 | | | | | | | |

## B. PROCESS WASTEWATER ANALYSES

### Lagoon Water

Sample and source description: Lagoon Water

Sample date: 02/20/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.40

| | Kjeldahl-N (mg/L) | NH4-N (mg/L) | NH3-N (mg/L) | Nitrate-N (mg/L) | Total P (mg/L) | Total K (mg/L) | Calcium (mg/L) | Magnes. (mg/L) | Sodium (mg/L) | Bicarb. (mg/L) | Carb. (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 224.00 | 199.00 | 0.00 | 0.70 | 78.70 | 472.00 | | | | | | | | 7,150.00 | 3,600 |
| **DL** | 1.00 | 0.50 | 0.50 | 0.10 | 0.10 | 0.50 | | | | | | | | 10.00 | 10 |

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 06/14/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.40

| | Kjeldahl-N (mg/L) | NH4-N (mg/L) | NH3-N (mg/L) | Nitrate-N (mg/L) | Total P (mg/L) | Total K (mg/L) | Calcium (mg/L) | Magnes. (mg/L) | Sodium (mg/L) | Bicarb. (mg/L) | Carb. (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 363.00 | 223.00 | 0.00 | 1.00 | 86.10 | 547.00 | | | | | | | | 8,170.00 | 4,110 |
| **DL** | 1.00 | 0.50 | 0.50 | 0.10 | 0.10 | 0.50 | | | | | | | | 0.01 | 10 |

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 09/25/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.40

| | Kjeldahl-N (mg/L) | NH4-N (mg/L) | NH3-N (mg/L) | Nitrate-N (mg/L) | Total P (mg/L) | Total K (mg/L) | Calcium (mg/L) | Magnes. (mg/L) | Sodium (mg/L) | Bicarb. (mg/L) | Carb. (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 7,110.00 | 132.00 | 0.00 | 0.30 | 71.50 | 529.00 | | | | | | | | 7,110.00 | 4,150 |
| **DL** | 1.00 | 0.50 | 0.50 | 0.10 | 0.10 | 0.50 | | | | | | | | 10.00 | 10 |

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 11/15/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.50

| | Kjeldahl-N (mg/L) | NH4-N (mg/L) | NH3-N (mg/L) | Nitrate-N (mg/L) | Total P (mg/L) | Total K (mg/L) | Calcium (mg/L) | Magnes. (mg/L) | Sodium (mg/L) | Bicarb. (mg/L) | Carb. (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 322.00 | 178.00 | 0.00 | 0.00 | 91.50 | 642.00 | | | | | | | | 7,990.00 | 4,950 |
| **DL** | 1.00 | 0.50 | 0.10 | 0.10 | 0.10 | 0.50 | | | | | | | | 10.00 | 10 |

## C. FRESH WATER ANALYSES

Domestic Well - 4-1 Facility

## Domestic Well - 4-1 Facility

### 4-1 Facility

Sample description: 4-1 Facility

Sample date: 11/13/2023 Source of analysis: Lab analysis

| | Total N (mg/L) | NH4-N (mg/L) | Nitrate-N (mg/L) | Calcium (mg/L) | Magnesium (mg/L) | Sodium (mg/L) | Bicarbonate (mg/L) | Carbonate (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 0.40 | | 0.40 | | | | | | | | 3,260.00 | |
| **DL** | 0.10 | | 0.10 | | | | | | | | 10.00 | |

## Domestic Well - 4-2 House

### 4-2 House

Sample description: 4-2 House

Sample date: 11/13/2023 Source of analysis: Lab analysis

| | Total N (mg/L) | NH4-N (mg/L) | Nitrate-N (mg/L) | Calcium (mg/L) | Magnesium (mg/L) | Sodium (mg/L) | Bicarbonate (mg/L) | Carbonate (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 4.20 | | 4.20 | | | | | | | | 2,560.00 | |
| **DL** | 0.10 | | 0.10 | | | | | | | | 10.00 | |

## Irrigation Water - Canal

### Canal

Sample description: Canal

Sample date: 07/17/2023 Source of analysis: Lab analysis

| | Total N (mg/L) | NH4-N (mg/L) | Nitrate-N (mg/L) | Calcium (mg/L) | Magnesium (mg/L) | Sodium (mg/L) | Bicarbonate (mg/L) | Carbonate (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 0.00 | | | | | | | | | | 32.00 | 33 |
| **DL** | 0.10 | | | | | | | | | | 10.00 | 10 |

**D. SOIL ANALYSES**

Field 2

Field 2

Sample and source description: Field 2

Sample date: 11/29/2023 Source of analysis: Lab analysis

| | Nitrate-N (mg/kg) | Total P (mg/kg) | Soluble P (mg/kg) | K (mg/kg) | EC (µmhos/cm) | Organic matter (%) | Total salt (mg/kg) |
|---|---|---|---|---|---|---|---|
| **Value** | | | 247.00 | | | | |
| **DL** | | | 2.00 | | | | |

Field 3

Field 3

Sample and source description: Field 3

Sample date: 01/29/2023 Source of analysis: Lab analysis

| | Nitrate-N (mg/kg) | Total P (mg/kg) | Soluble P (mg/kg) | K (mg/kg) | EC (µmhos/cm) | Organic matter (%) | Total salt (mg/kg) |
|---|---|---|---|---|---|---|---|
| **Value** | | | 117.00 | | | | |
| **DL** | | | 2.00 | | | | |

West 20

West 20

Sample and source description: West 20

Sample date: 11/29/2023 Source of analysis: Lab analysis

| | Nitrate-N (mg/kg) | Total P (mg/kg) | Soluble P (mg/kg) | K (mg/kg) | EC (µmhos/cm) | Organic matter (%) | Total salt (mg/kg) |
|---|---|---|---|---|---|---|---|
| **Value** | | | 153.00 | | | | |
| **DL** | | | 2.00 | | | | |

## E. PLANT TISSUE ANALYSES

Field 2 - 11/15/2022: Oats, silage-soft dough

## Field 2 - 11/15/2022: Oats, silage-soft dough

### Fields 2 & 3

Sample and source description: Fields 2 & 3

Sample date: 05/05/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 80.3 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 17,400.00 | 3,100.00 | 24,800.00 | | 11.40 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

## Field 2 - 05/28/2023: Corn, silage

### Fields 2 & 3

Sample and source description: Fields 2 & 3

Sample date: 09/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.7 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 16,800.00 | 3,500.00 | 16,200.00 | | 6.70 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

## Field 3 - 11/15/2022: Oats, silage-soft dough

### Fields 2 & 3

Sample and source description: Fields 2 & 3

Sample date: 05/05/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 80.3 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 17,400.00 | 3,100.00 | 24,800.00 | | 11.40 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

## Field 3 - 05/28/2023: Corn, silage

### Fields 2 & 3

Sample and source description: Fields 2 & 3

Sample date: 09/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.7 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 16,800.00 | 3,500.00 | 16,200.00 | | 6.70 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

## West 20 - 11/15/2022: Oats, silage-soft dough

### West 20

Sample and source description: West 20

Sample date: 05/05/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 79.7 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 17,100.00 | 2,900.00 | 25,500.00 | | 2.30 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

## West 20 - 05/28/2023: Corn, silage

### West 20

Sample and source description: West 20

Sample date: 09/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.3 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 17,700.00 | 3,500.00 | 16,300.00 | | 6.70 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.01 |

## F. SUBSURFACE (TILE) DRAINAGE ANALYSES

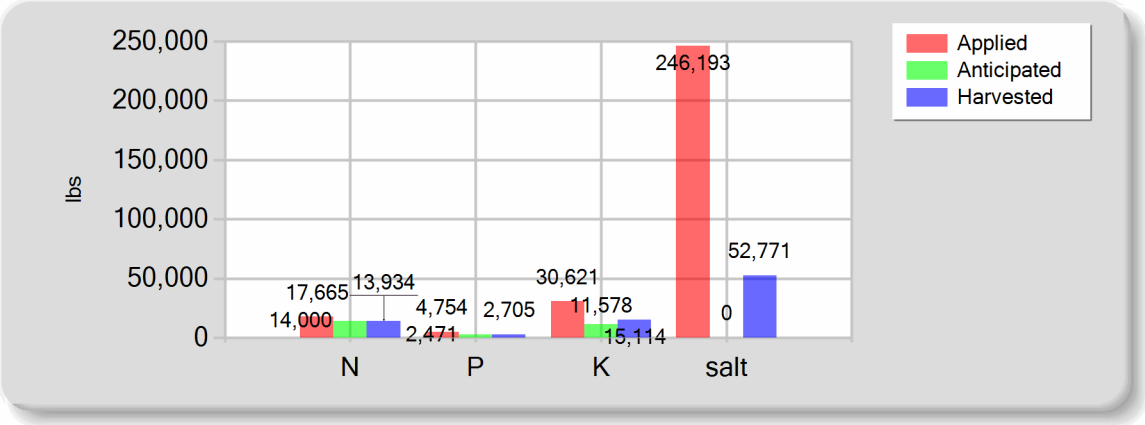
*No subsurface (tile) drainage analyses entered.*

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

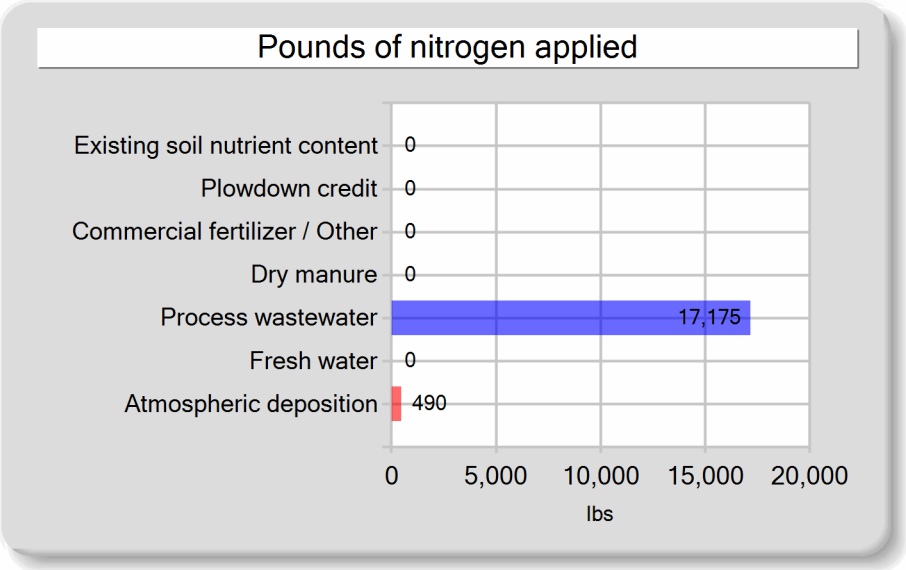
## A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

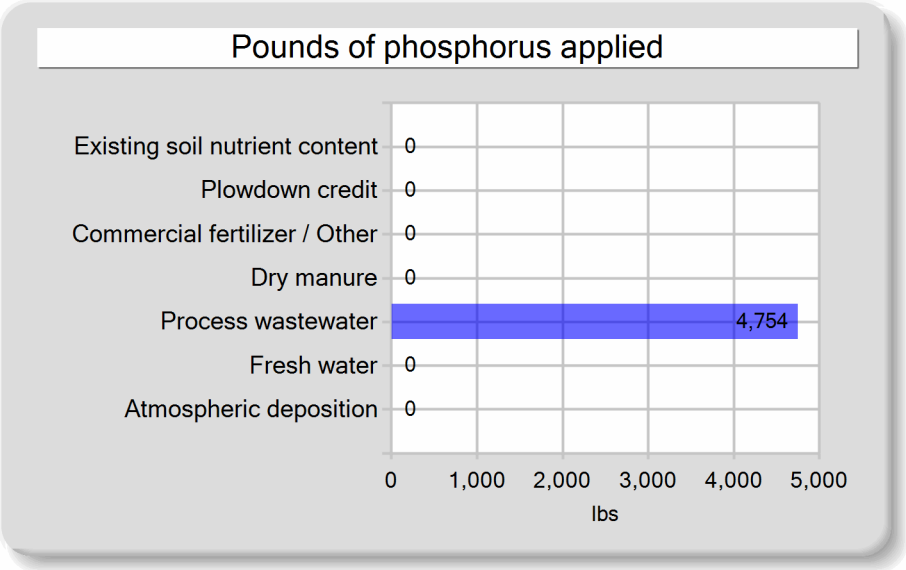
| | Total N (lbs) | Total P (lbs) | Total K (lbs) | Total salt (lbs) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 17,175.00 | 4,753.90 | 30,620.55 | 232,949.94 |
| Fresh water | 0.00 | 0.00 | 0.00 | 13,242.69 |
| Atmospheric deposition | 490.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 17,665.00 | 4,753.90 | 30,620.55 | 246,192.63 |
| Anticipated crop nutrient removal | 14,000.00 | 2,471.00 | 11,578.00 | 0.00 |
| Actual crop nutrient removal | 13,933.72 | 2,705.11 | 15,114.30 | 52,771.33 |
| Nutrient balance | 3,731.29 | 2,048.79 | 15,506.25 | 193,421.30 |
| Applied to removed ratio | 1.27 | 1.76 | 2.03 | 4.67 |

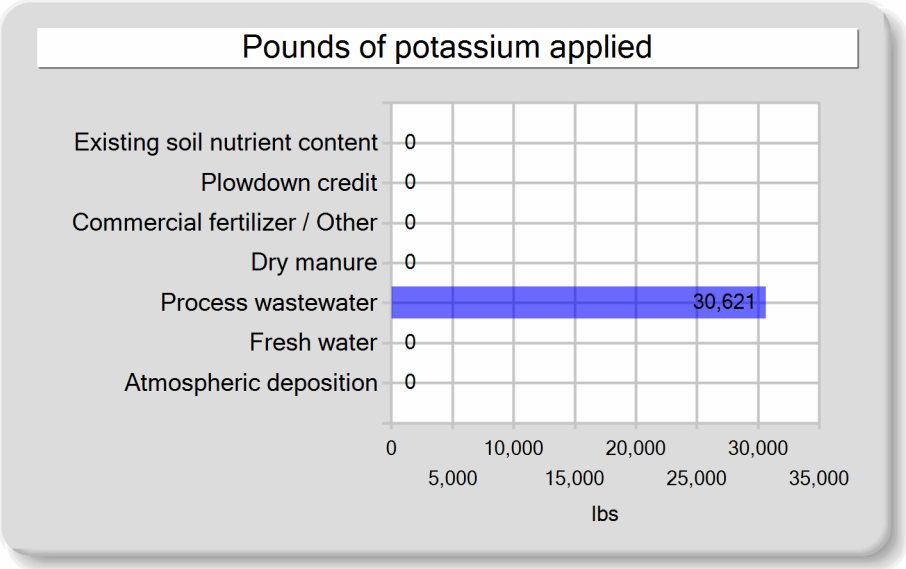
## B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

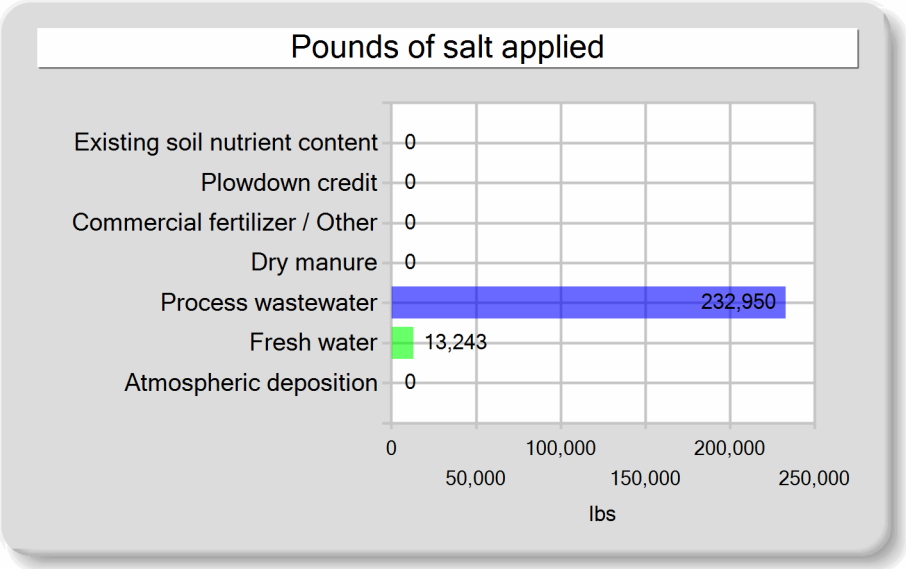


**C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE**



## EXCEPTION REPORTING

### A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

*No manure or process wastewater discharges occurred during the reporting period.*

### B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

*No stormwater discharges occurred during the reporting period.*

### C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

*No land application area to surface water discharges occurred during the reporting period.*

## NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

### A. NUTRIENT MANAGEMENT PLAN STATEMENTS

| | |
|---|---|
| Was the facility's NMP updated in the reporting period? | Yes |
| Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order? | Yes |
| Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order? | Yes |

### B. EXPORT AGREEMENT STATEMENT

| | |
|---|---|
| Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period? | No |

ADDITIONAL NOTES

**A. NOTES**

*No notes entered for this annual report.*

CERTIFICATION

**A. OWNER AND/OR OPERATOR CERTIFICATION**

*I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.*

| | |
|---|---|
| Scott Sanders<br>Scott Sanders (Jun 18, 2024 16:30 PDT) | Cathy Sanders |
| SIGNATURE OF OWNER OF FACILITY | SIGNATURE OF OPERATOR OF FACILITY |
| Scott & Sharlee Sanders | Cathy Sanders |
| PRINT OR TYPE NAME | PRINT OR TYPE NAME |
| 06/18/2024 | 06/18/2024 |
| DATE | DATE |

## ATTACHMENT D

## Manure Tracking Manifest
## For
## Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

**Operator Information**:

Name of Operator: **Cathy or Scott Sanders**

Name of Dairy Facility: **Fire-Lake Jerseys**

Facility Address: **11301 W. Bradbury Rd, Turlock, CA 95380**

Contact Person Name and Phone Number: **Cathy Sanders (209) 632-2924**

**Manure Hauler Information:**

Name of Hauling Company/Person: **Joshua Holsapple's Fertilizer Spreading, Inc.**

Address of Hauling Company /Person: **400 N. Central Avenue, Turlock, CA 95380**

Contact Person and Phone Number: **Joshua Holsapple (209) 668-4727**

**Destination Information:**

Composting Facility / Broker / Farmer / Other (identify) **Broker** (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

| Name | Number and Street | City | Zip Code | Phone Number |
|---|---|---|---|---|
| J.H.F.S. Inc. | 400 N. Central Avenue | Turlock | 95380 | (209) 668-4727 |

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

| Number and Street | City | Zip Code | Assessor's Parcel Number |
|---|---|---|---|
| 400 N. Central Avenue | Turlock | 95380 | |

Dates Hauled: **5/24/2023**

**Amount Hauled:**

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: **900** Tons

Manure Solids Content (if amount reported in tons): **91.3%**

Method used to determine amount of manure: _____

**Number of Truckloads x Average Truck Weight**

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: **N/A** Gallons

Method used to determine volume of process wastewater: **N/A**

**Written Agreement:**

****NOT NECESSARY FOR SOLID MANURE MANIFEST****

**Certification**:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Cathy Sanders (Jun 18, 2024 16:14 PDT) Date: 06/18/2024

Hauler's Signature: _____ Date: 5/24/2023

DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Fire-Lake Jerseys

Received: 11/14/2023 15:45
Reported: 11/21/2023 09:23

## Samples in this Report

| Lab ID | Sample | Matrix | Sampled By | Crop | Date Sampled |
|---|---|---|---|---|---|
| 23K0673-01 | 4-1 Facility | Ag Water | T. Madruga | | 11/13/2023 15:30 |
| 23K0673-02 | 4-2 House | Ag Water | T. Madruga | | 11/13/2023 16:00 |

Default Cooler Temperature on Receipt °C: 3.4
Containers Intact
COC/Labels Agree
Received On Ice

## Notes and Definitions

| Item | Definition |
|---|---|
| H | Hold Time Exceeded |
| MCL | Drinking Water Maximum Contaminant Level |
| ND | Analyte NOT DETECTED at or above the reporting limit. |
| NES | Not Enough Sample |
| * | Not Taken |
| RPD | Relative Percent Difference |
| %REC | Percent Recovery |
| Source | Sample that was matrix spiked or duplicated. |

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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8119 Ellington Ave
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Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Fire-Lake Jerseys

Received: 11/14/2023 15:45
Reported: 11/21/2023 09:23

## Sample Results

**Sample: 4-1 Facility**
**23K0673-01 (Water)**

Sampled: 11/13/2023 15:30
Sampled By: T. Madruga

| Analyte | Result | Units | Reporting Limit | DIL | DW MCL | Date/Time Analyzed | Method | Notes | Batch |
|---|---|---|---|---|---|---|---|---|---|
| **Electrical Conductivity** | **3.26** | mmhos/cm | 0.01 | 1 | | 11/16/23 12:51 | SM 2510 B | | BEK0477 |
| **Electrical Conductivity umhos** | **3260** | umhos/cm | 10.0 | 1 | | 11/16/23 12:51 | SM 2510 B | | BEK0477 |
| Ammonia (as N) | ND | mg/L | 0.00 | 1 | | 11/13/23 15:30 | Field | | BEK0419 |
| **Nitrate Nitrogen as NO3N** | **0.4** | mg/L | 0.1 | 1 | 10 | 11/14/23 21:40 | EPA 300.0 | | BEK0412 |
| **Temperature** | **25.0** | units | 0.0 | 1 | | 11/16/23 12:51 | SM 4500-H+ | H | BEK0477 |
| **pH** | **7.8** | units | 1.0 | 1 | | 11/16/23 12:51 | SM 4500-H+ | H | BEK0477 |

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8119 Ellington Ave
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Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Fire-Lake Jerseys

Received: 11/14/2023 15:45
Reported: 11/21/2023 09:23

## Sample Results
**(Continued)**

**Sample: 4-2 House**
**23K0673-02 (Water)**

Sampled: 11/13/2023 16:00
Sampled By: T. Madruga

| Analyte | Result | Units | Reporting Limit | DIL | DW MCL | Date/Time Analyzed | Method | Notes | Batch |
|---|---|---|---|---|---|---|---|---|---|
| **Electrical Conductivity** | **2.56** | mmhos/cm | 0.01 | 1 | | 11/16/23 12:52 | SM 2510 B | | BEK0477 |
| **Electrical Conductivity umhos** | **2560** | umhos/cm | 10.0 | 1 | | 11/16/23 12:52 | SM 2510 B | | BEK0477 |
| Ammonia (as N) | ND | mg/L | 0.00 | 1 | | 11/13/23 16:00 | Field | | BEK0419 |
| **Nitrate Nitrogen as NO3N** | **4.2** | mg/L | 0.1 | 1 | 10 | 11/15/23 00:20 | EPA 300.0 | | BEK0412 |
| **Temperature** | **25.0** | units | 0.0 | 1 | | 11/16/23 12:52 | SM 4500-H+ | H | BEK0477 |
| **pH** | **7.8** | units | 1.0 | 1 | | 11/16/23 12:52 | SM 4500-H+ | H | BEK0477 |

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LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Fire-Lake Jerseys

Received: 11/14/2023 15:45
Reported: 11/21/2023 09:23

## Quality Control

| Analyte | Result | Qual | Reporting Limit | Units | Spike Level | Source Result | %REC | %REC Limits | RPD | RPD Limit |
|---|---|---|---|---|---|---|---|---|---|---|
| ***Batch: BEK0412*** | | | | | | | | | | |
| **Blank (BEK0412-BLK1)** | | | | | Prepared & Analyzed: 11/14/2023 | | | | | |
| Nitrate Nitrogen as NO3N | ND | | 0.1 | mg/L | | | | | | |
| **Blank (BEK0412-BLK2)** | | | | | Prepared & Analyzed: 11/14/2023 | | | | | |
| Nitrate Nitrogen as NO3N | ND | | 0.1 | mg/L | | | | | | |
| **Blank (BEK0412-BLK3)** | | | | | Prepared: 11/14/2023 Analyzed: 11/15/2023 | | | | | |
| Nitrate Nitrogen as NO3N | ND | | 0.1 | mg/L | | | | | | |
| **LCS (BEK0412-BS1)** | | | | | Prepared & Analyzed: 11/14/2023 | | | | | |
| Nitrate Nitrogen as NO3N | 5.0 | | 0.1 | mg/L | 5.000 | | 100 | 90-110 | | |
| **LCS (BEK0412-BS2)** | | | | | Prepared: 11/14/2023 Analyzed: 11/15/2023 | | | | | |
| Nitrate Nitrogen as NO3N | 5.0 | | 0.1 | mg/L | 5.000 | | 100 | 90-110 | | |
| **Duplicate (BEK0412-DUP1)** | | **Source: 23K0676-01** | | | Prepared & Analyzed: 11/14/2023 | | | | | |
| Nitrate Nitrogen as NO3N | 15.7 | | 0.1 | mg/L | | 15.5 | | | 0.802 | 10 |
| **Duplicate (BEK0412-DUP2)** | | **Source: 23K0676-02** | | | Prepared: 11/14/2023 Analyzed: 11/15/2023 | | | | | |
| Nitrate Nitrogen as NO3N | 36.4 | | 0.1 | mg/L | | 36.4 | | | 0.0192 | 10 |
| **Matrix Spike (BEK0412-MS1)** | | **Source: 23K0676-01** | | | Prepared & Analyzed: 11/14/2023 | | | | | |
| Nitrate Nitrogen as NO3N | 20.6 | | 0.1 | mg/L | 5.000 | 15.5 | 102 | 90-110 | | |
| **Matrix Spike (BEK0412-MS2)** | | **Source: 23K0676-02** | | | Prepared: 11/14/2023 Analyzed: 11/15/2023 | | | | | |
| Nitrate Nitrogen as NO3N | 42.1 | | 0.1 | mg/L | 5.000 | 36.4 | 113 | 90-110 | | |
| **Reference (BEK0412-SRM1)** | | | | | Prepared & Analyzed: 11/14/2023 | | | | | |
| Nitrate Nitrogen as NO3N | 9.8 | | | mg/L | 10.00 | | 97.9 | 90-110 | | |
| **Reference (BEK0412-SRM2)** | | | | | Prepared: 11/14/2023 Analyzed: 11/15/2023 | | | | | |
| Nitrate Nitrogen as NO3N | 9.9 | | | mg/L | 10.00 | | 98.7 | 90-110 | | |
| **Reference (BEK0412-SRM3)** | | | | | Prepared: 11/14/2023 Analyzed: 11/15/2023 | | | | | |
| Nitrate Nitrogen as NO3N | 9.9 | | | mg/L | 10.00 | | 99.2 | 90-110 | | |

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DELLAVALLE™ LABORATORY INC

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8119 Ellington Ave
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Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Fire-Lake Jerseys

Received: 11/14/2023 15:45
Reported: 11/21/2023 09:23

## Quality Control
**(Continued)**

| Analyte | Result | Qual | Reporting Limit | Units | Spike Level | Source Result | %REC | %REC Limits | RPD | RPD Limit |
|---|---|---|---|---|---|---|---|---|---|---|
| ***Batch: BEK0477*** | | | | | | | | | | |
| **Blank (BEK0477-BLK1)** | | | | | Prepared: 11/15/2023 | Analyzed: 11/16/2023 | | | | |
| Temperature | 25.0 | | 0.0 | units | | | | | | |
| Electrical Conductivity | ND | | 0.01 | mmhos/cm | | | | | | |
| Electrical Conductivity umhos | ND | | 10.0 | umhos/cm | | | | | | |
| pH | 5.5 | | 1.0 | units | | | | | | |
| **Blank (BEK0477-BLK2)** | | | | | Prepared: 11/15/2023 | Analyzed: 11/16/2023 | | | | |
| Electrical Conductivity | ND | | 0.01 | mmhos/cm | | | | | | |
| Temperature | 25.0 | | 0.0 | units | | | | | | |
| Electrical Conductivity umhos | ND | | 10.0 | umhos/cm | | | | | | |
| pH | 7.5 | | 1.0 | units | | | | | | |
| **Blank (BEK0477-BLK3)** | | | | | Prepared: 11/15/2023 | Analyzed: 11/16/2023 | | | | |
| Electrical Conductivity | ND | | 0.01 | mmhos/cm | | | | | | |
| Temperature | 25.0 | | 0.0 | units | | | | | | |
| Electrical Conductivity umhos | ND | | 10.0 | umhos/cm | | | | | | |
| pH | 7.3 | | 1.0 | units | | | | | | |
| **Duplicate (BEK0477-DUP1)** | | **Source: 23K0667-01** | | | Prepared: 11/15/2023 | Analyzed: 11/16/2023 | | | | |
| Electrical Conductivity | 4.04 | | 0.01 | mmhos/cm | | 4.04 | | | 0.0495 | 10 |
| pH | 7.8 | | 1.0 | units | | 7.6 | | | 1.82 | 10 |
| Electrical Conductivity umhos | 4040 | | 10.0 | umhos/cm | | 4040 | | | 0.0495 | 10 |
| **Duplicate (BEK0477-DUP2)** | | **Source: 23K0676-03** | | | Prepared: 11/15/2023 | Analyzed: 11/16/2023 | | | | |
| Electrical Conductivity | 1.35 | | 0.01 | mmhos/cm | | 1.36 | | | 0.791 | 10 |
| Electrical Conductivity umhos | 1350 | | 10.0 | umhos/cm | | 1360 | | | 0.791 | 10 |
| pH | 7.6 | | 1.0 | units | | 7.6 | | | 0.264 | 10 |
| **Reference (BEK0477-SRM1)** | | | | | Prepared: 11/15/2023 | Analyzed: 11/16/2023 | | | | |
| Electrical Conductivity | 578 | | | umhos/cm | 538.0 | | 107 | 90-110 | | |
| **Reference (BEK0477-SRM2)** | | | | | Prepared: 11/15/2023 | Analyzed: 11/16/2023 | | | | |
| pH | 7.5 | | | units | 7.520 | | 100 | 67021-101.32 | | |
| **Reference (BEK0477-SRM3)** | | | | | Prepared: 11/15/2023 | Analyzed: 11/16/2023 | | | | |
| Electrical Conductivity | 1060 | | | umhos/cm | 1000 | | 106 | 90-110 | | |
| Electrical Conductivity umhos | 1060 | | | umhos/cm | 1000 | | 106 | 90-110 | | |
| **Reference (BEK0477-SRM4)** | | | | | Prepared: 11/15/2023 | Analyzed: 11/16/2023 | | | | |
| Electrical Conductivity | 1080 | | | umhos/cm | 1000 | | 108 | 90-110 | | |
| Electrical Conductivity umhos | 1080 | | | umhos/cm | 1000 | | 108 | 90-110 | | |
| **Reference (BEK0477-SRM5)** | | | | | Prepared: 11/15/2023 | Analyzed: 11/16/2023 | | | | |
| Electrical Conductivity | 1060 | | | umhos/cm | 1000 | | 106 | 90-110 | | |

*The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.*

DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Fire-Lake Jerseys

Received: 11/14/2023 15:45
Reported: 11/21/2023 09:23

## Quality Control
**(Continued)**

| Analyte | Result | Qual | Reporting Limit | Units | Spike Level | Source Result | %REC | %REC Limits | RPD | RPD Limit |
|---|---|---|---|---|---|---|---|---|---|---|
| ***Batch: BEK0477 (Continued)*** | | | | | | | | | | |
| **Reference (BEK0477-SRM5)** | | | | | Prepared: 11/15/2023 | Analyzed: 11/16/2023 | | | | |
| Electrical Conductivity umhos | 1060 | | | umhos/cm | 1000 | | 106 | 90-110 | | |
| **Reference (BEK0477-SRM6)** | | | | | Prepared: 11/15/2023 | Analyzed: 11/16/2023 | | | | |
| pH | 4.0 | | | units | 4.000 | | 100 | 97.5-102.5 | | |
| **Reference (BEK0477-SRM7)** | | | | | Prepared: 11/15/2023 | Analyzed: 11/16/2023 | | | | |
| pH | 4.0 | | | units | 4.000 | | 100 | 97.5-102.5 | | |
| **Reference (BEK0477-SRM8)** | | | | | Prepared: 11/15/2023 | Analyzed: 11/16/2023 | | | | |
| pH | 4.0 | | | units | 4.000 | | 100 | 97.5-102.5 | | |

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11/14/23 15:45 23K0673

# DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 · 800 228-9896 · Fax 559 268-8174

JG No. Samples 2 No. of Bottles: 2

Purchase Order No | Bill To: Acct # 19840 | Cons # 08

Results Need By

Water Type: [ ] Drinking Water [ ] Wastewater
[X] Ag Water [ ] Groundwater [ ] Monitoring Well

**Name:** R & J Avila Inc
**Address:** 8119 Ellington Avenue
**City:** Hilmar **State:** CA **Zip:** 95324
**Telephone:** (209) 669-7919 **Fax:** (209) 634-4604
**Cell/Email:** 209 678-3086 Rui / 209 678-1723 John frontdesk.randjavila@gmail.com

**COPY TO:**

**REQUESTED BY:** Rui & John Avila
**PROJECT** Fire-Lake Jerseys
**ID CROP:** N/A

[X] Copy of Chain [X] QA/QC Documents

**Sampled By:** T. Madruga

**Other:**

**Analysis and Bottles Required: (Please indicate Analysis)**

**( X ) DWW1: EC, pH, $NO_3$-N (NH4-N Field Test)**
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DWW2: DWW1 Plus $SO_4$, $CO_3$, $HCO_3$, Cl, Ca, Mg, Na, TDS
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DCW1: EC, $NO_3$-N, TDS + TKN & TN
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW1: EC, pH, $NO_3$-N, $NH_4$-N, TKN, TDS, TP, TK
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW2: DPW1 Plus $SO_4$, $CO_3$, $HCO_3$, Cl, Ca, Mg, Na
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) Other

| | Description of Samples | Date Sampled | Time Sampled | Rec'd Temp °C | Field $NH_4$-N |
|---|---|---|---|---|---|
| 1 | 4-1 Facility | 11/13/2023 | 1530 | 3.4 | ND |
| 2 | 4-2 House | 11/13/2023 | 1600 | 3.6 | ND |
| 3 | | | | | |
| 4 | | | | | |
| 5 | | | | | |
| 6 | | | | | |
| 7 | | | | | |
| 8 | | | | | |
| 9 | | | | | |
| 10 | | | | | |

IR Thermometer SN: 192603727
Correction Factor: 0°C
Calibration Due: 12/22/2023
Location: Laboratory

**CHAIN OF CUSTODY**

| Carrier | Signature | Company | Received (Date/Time) | Relinquished (Date/Time) |
|---|---|---|---|---|
| First | John Avila | R & J Avila, Inc. | 11/13/2023 @ 1600 | 11/14/2023 @ 900 |
| Second | JLF | Dellavalle Laboratory, Inc. | 11/14/2023 @ 900 | 11-14-23 3:30 |
| Third | | | | |
| Fourth | | | 11/14 15:45 | |

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.
If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (*cal* ). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through ***cal*** under its Rules and Procedures. The parties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration,reasonable attorneys' fees of Dellavalle Laboratory.

***Invoicing Information:***
Sampling hrs
Miles
Consulting

***Shipping***
$ In
$ Out

| Amt Paid | Rec By | Check # | Date |
|---|---|---|---|
| | | | |

**Signature**

Samples recieved in cooler with ice (coolant)
[ ] Yes [ ] No

11/14/23 15:45 23K0673

**Shipping Information:** Fed Ex ☐ UPS ☐ GSO ☐ Picked-Up ☒ Walk In ☐ DLI Sampler ☐
Ontrac ☐ PMS ☐ USPS ☐ Other ☐ _____

**Shipping Container:** Ice Chest ☒ None ☐ Box ☐

**Refrigerant:** Wet Ice: ☒ Blue Ice: ☐ None: ☐

| Container(s) Received | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| **Plastics:** | | | | | | | | | | |
| 100 mL sterile plastic $Na_2S_2O_3$ (Green) | | | | | | | | | | |
| 250 mL unpreserved (White) | | | | | | | | | | |
| 250 mL $HNO_3$ (Red) | | | | | | | | | | |
| 250 mL $H_2SO_4$ (Yellow) | | | | | | | | | | |
| 500 mL unpreserved (White) | | | | | | | | | | |
| 500 mL $HNO_3$ (Red) | | | | | | | | | | |
| 500 mL $H_2SO_4$ (Yellow) | | | | | | | | | | |
| 1 L unpreserved (White) | 1 | 1 | | | | | | | | |
| 1 L unpreserved (BOD) (Purple) | | | | | | | | | | |
| 1 L $HNO_3$ (Red) | | | | | | | | | | |
| 1 L $H_2SO_4$ (Yellow) | | | | | | | | | | |
| **VOA Vials in sets of 3:** | | | | | | | | | | |
| 40 mL VOA, HCl (Blue) | | | | | | | | | | |
| 40 mL VOA, $Na_2S_2O_3$ (Green) | | | | | | | | | | |
| 40 mL VOA, unpreserved (White) | | | | | | | | | | |
| **Glass:** | | | | | | | | | | |
| 250 mL AG unpreserved (White) | | | | | | | | | | |
| 250 mL AG $H_2SO_4$ (Yellow) | | | | | | | | | | |
| 250 mL AG $Na_2S_2O_3$ (Green) | | | | | | | | | | |
| 250 mL AG $Na_2S_2O_3$ + MCAA | | | | | | | | | | |
| 500 mL glass unpreserved (White) | | | | | | | | | | |
| 500 mL AG HCl (Blue) | | | | | | | | | | |
| 1 L AG unpreserved (White) | | | | | | | | | | |
| 1 L AG $H_2SO_4$ (Yellow) | | | | | | | | | | |
| 1 L AG $Na_2S_2O_3$ (Green) | | | | | | | | | | |
| 1 L AG HCl (Blue) | | | | | | | | | | |
| **Special:** | | | | | | | | | | |
| $Cr^{6+}$ w/Borate/HCO3/CO3 | | | | | | | | | | |
| Cyanide - 500 mL NaOH | | | | | | | | | | |
| Asbestos - 2 - 1L P wrapped in foil | | | | | | | | | | |
| 500 mL or 1 L **AP** $Na_2S_2O_3$ (Green) | | | | | | | | | | |
| 1 L **AG** or **P** NaOH + ZnAc (Sulfide) | | | | | | | | | | |
| Chlorite/Bromate 250 mL AG with EDA | | | | | | | | | | |
| **TTHM** | | | | | | | | | | |
| **HAA5** | | | | | | | | | | |
| 40 mL VOA, $Na_2S_2O_3$ + MCAA | | | | | | | | | | |